Vipul Sapra

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EDUCATION

University of Cincinnati, Carl H. Lindner College of Business — **Cincinnati, OH**
Master of Science in Information Systems and Graduate Certificate in Data Science — **April 2022**
Cumulative GPA: 3.95 / 4.0

Pune University, Maharashtra Institute of Technology — **Pune, India**
Bachelor of Engineering in Petrochemical — **June 2017**

TECHNICAL SKILLS

Concepts: Agentic AI, Generative AI, LLM Orchestration, Autonomous Anomaly Detection, Lakehouse Architecture, Real-Time Streaming & CDC, Data Governance & Lineage, Predictive Modeling & Time-Series Forecasting, MLOps, CI/CD Automation

Programming Languages & Tools: Python, PySpark, Databricks, Snowflake, Azure (ADF, Databricks, Synapse, Azure SQL), AWS (S3, Redshift), Power BI (DAX), Tableau, Collibra, XGBoost, Scikit-Learn, SARIMA, GitHub Copilot, KIRO IDE

Certifications: Data Science Graduate Certificate – University of Cincinnati

PROFESSIONAL EXPERIENCE

Charter Communications — Charlotte, NC

Senior Data & AI Systems Engineering Technical Leader — **May 2022 – Present**

- **Architected and deployed custom autonomous agentic workflows** (KIRO IDE, prompt orchestration, GitHub Copilot) to automate schema discovery, data profiling, regex classification, and code migration, reducing manual delivery turnaround by 40-50%.
- **Engineered an autonomous machine learning anomaly detection framework** identifying schema variations, distribution drift, and ETL discrepancies with 95% precision, safeguarding downstream BI and pipeline integrity.
- **Built and deployed high-throughput cloud lakehouse data platforms** using Azure (ADF, Databricks, Synapse, Azure SQL), AWS (S3, Redshift), Snowflake, and PySpark to crunch hundreds of terabytes with 99.9% data availability SLAs.
- **Designed robust batch and streaming ETL/ELT pipelines** implementing dimensional models (Star/Snowflake schemas, SCD Type 1/2) and event-driven data flows to power real-time analytics and downstream AI/ML consumption.
- **Engineered automated data quality and source-to-target reconciliation firewalls** integrated with Collibra lineage tracking to establish enterprise-grade governance, reducing operational turnaround by over 80%.

Data Scientist & Predictive Machine Learning Specialist — **May 2022 – Present**

- **Developed predictive machine learning models at scale** (XGBoost, Random Forests, Scikit-Learn) forecasting customer behavior and delivering actionable operational signals that improved service reliability by 20%.
- **Architected advanced time-series forecasting models** (SARIMA) to predict truck rolls and field support demand, optimizing workforce capacity allocation and reducing response times by 20%.
- **Engineered automated testing and evaluation frameworks for conversational AI** (SpectrumIQ enterprise virtual assistant), cutting manual QA turnaround by 50% and accelerating production release cycles.

Business Intelligence & Decision Analytics Specialist — **May 2022 – Present**

- **Designed and deployed executive Key Operating Metric (KOM) dashboards** and semantic models in Power BI (DAX) and Tableau, translating complex transactional telemetry into strategic corporate decisions.
- **Translated complex transactional and operational telemetry** into high-impact self-service reporting assets and executive scorecards aligning with core business priorities.
- **Mentored engineering pods and drove technical consensus** across matrixed teams to establish rigorous standards for data modeling, semantic layers, and operational observability.

Ameritas Insurance — Cincinnati, OH

Data Governance Intern — **September 2021 – April 2022**

- **Co-engineered the enterprise-wide Data Advocacy & Governance Program** across insurance portfolios, establishing foundational frameworks for data democratization and organizational accountability.
- **Architected a comprehensive Data Intelligence ecosystem in Collibra**, establishing a centralized Data Dictionary, automated Data Quality rules, and end-to-end **Data Lineage** to ensure regulatory compliance and transparency for financial and insurance assets.

Accenture — Pune, India

Test Engineering Analyst — **September 2017 – December 2020**

- **Orchestrated end-to-end Quality Assurance and system integration testing** for mission-critical client applications at **Allianz Insurance**, leveraging Agile SDLC methodologies to ensure rapid, zero-defect feature deployments.

- **Spearheaded the Health Insurance module QA strategy**, directing cross-functional efforts for regression testing, test data validation, and the seamless rollout of complex new business functionalities.
- **Drove operational excellence by automating core regression test suites**, significantly increasing engineering productivity and reducing manual testing overhead through scalable automation frameworks.